

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has lately emerged as a compelling paradigm for augmenting the reasoning faculties of large language models. Yet prevailing instantiations of this paradigm, GRPO foremost among them, are persistently afflicted by the pathology of vanishing gradients. This deficiency is attributable in the main to two interacting mechanisms: the imposition of fixed clipping bounds upon token-level probability ratios, and the standardization of rewards that happen to be identical—jointly precipitating ineffectual gradient updates and a pronounced underexploitation of the responses the policy has already generated. To remedy these shortcomings, we advance **Dynamic Clipping Policy Optimization (DCPO)**, a formulation predicated on two complementary devices: a dynamic clipping strategy that adaptively recalibrates the clipping bounds in accordance with token-specific prior probabilities, thereby amplifying token-level exploration; and a smooth advantage-standardization technique that normalizes rewards over cumulative training steps, so as to elevate the response-level utilization of the generated corpus. Across four benchmarks and four distinct backbone models, DCPO attains state-of-the-art performance. In particular, on the AIME24 benchmark instantiated over the Qwen2.5-Math-7B model, DCPO registers an Avg@1 of 46.7 under greedy decoding together with an Avg@32 of 38.8 under 32-fold sampling, thereby eclipsing DAPO (36.7/31.6), GRPO (36.7/32.1) and GSPO (40.0/34.9). On the AIME25 benchmark founded upon Qwen2.5-14B, DCPO secures a performance of (23.3/19.0), likewise surpassing GRPO (13.3/10.5), DAPO (20.0/15.3) and GSPO (16.7/9.9). Beyond raw accuracy, DCPO delivers an average 28% enlargement of the nonzero-advantage fraction relative to GRPO across the four models, doubles the training efficiency attained by DAPO, and curtails the token clipping ratio by a full order of magnitude relative to both GRPO and DAPO, all while retaining its performance advantage. Taken together, these findings underscore the efficacy with which DCPO marshals generated data toward more sample-efficient reinforcement learning in large language models.

1 INTRODUCTION

Over the past several years, the reasoning capabilities of large language models (LLMs) have attracted an ever-intensifying degree of scholarly attention (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Within this landscape, Reinforcement Learning from Verifiable Rewards (RLVR) has established itself as a particularly promising avenue for the cultivation of such capabilities. By exploiting rule-based outcome rewards as its optimization signal, the method sharpens the competence of LLMs precisely in those domains that demand reflective, multi-step deliberation—mathematics and coding being the canonical exemplars (DeepSeek-AI et al., 2024).

The operationalization of RLVR is largely owed to the GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms. Although the robustness of GRPO has been corroborated across a considerable body of empirical work, more recent inquiry—DAPO (Yu et al., 2025) being a salient instance—has laid bare a number of critical infirmities. In particular, GRPO is prone to entropy collapse, a phenomenon manifested by the progressive decay of advantage values toward zero in the later stages of training. Compounding this, ill-calibrated weight clipping tends to expunge the learning signals borne by infrequently sampled tokens, thereby unduly constricting the diversity of the model’s exploratory behavior.

Confronting these difficulties, DAPO, for instance, advances the Clip-Higher strategy—which attenuates entropy collapse by loosening the upper clipping bound—together with a Dynamic Sampling strategy that culls samples exhibiting uniform rewards, thereby guaranteeing that every batch is populated exclusively by samples carrying valid gradients. These remedies, however, remain encumbered by two conspicuous limitations. (i) **Response-Level Inefficiency**: dynamic sampling, by its very construction, depresses sampling efficiency and thus retards the convergence of training. (ii) **Token Clipping Deficit**: our own analysis discloses that the optimal clipping bound is far from uniform across tokens, rendering any single, globally imposed upper bound inherently suboptimal. Even among approaches that studiously refrain from discarding zero-gradient responses outright, these limitations continue to persist.

In the present work we introduce **Dynamic Clipping Policy Optimization (DCPO)**, a novel methodology conceived expressly to redress the principal limitations enumerated above. First, we devise a **smoothing technique that standardizes advantages across cumulative steps**, thereby countering the gradient-attenuation problem that arises when the advantages of identically rewarded responses are nullified. By amalgamating the reward distribution of the entire body of cumulatively generated responses with that of the currently generated batch, this device affords markedly more efficient response-level exploration (see Section 2). Second, we propose a **dynamic clipping method that adaptively recalibrates the clipping bounds in light of the old, token-specific probabilities**, so as to overcome the drawbacks intrinsic to fixed bounds rather than merely substituting one pair of static upper and lower bounds for another. Concretely, tokens endowed with lower prior probabilities are accorded correspondingly wider clipping bounds, thereby opening up additional latitude for token-level exploration (see Section 2 and Appendix A.6). Finally, DCPO computes the mean loss solely over the tokens internal to each individual response, in preference to averaging across an entire batch; this design preserves the relative advantage structure obtaining among responses to a common prompt and forestalls the dilution effect that batch-level averaging would otherwise induce.

To furnish an empirical foundation, we constituted our training corpus by merging Math-DAPO-17k (Yu et al., 2025) with levels 3–5 of the MATH dataset (Liu et al., 2025), and we designated four widely used mathematical-reasoning benchmarks as the arena for evaluation. Experiments were conducted across four distinct models, with GRPO, DAPO, and—wherever available—GSPO serving as baselines. The evidence so obtained demonstrates that DCPO consistently secures the superior outcome, with especially pronounced gains accruing on the demanding AIME benchmarks: under 32-fold sampling on AIME24-Avg@32, DCPO-7B attains 38.8 as against 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), while on AIME25-Avg@32, DCPO-14B reaches 19.0 as against 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). Moreover, DCPO enlarges the average nonzero-advantage ratio by 28% relative to GRPO, doubles the training efficiency exhibited by DAPO, and compresses token clipping by an order of magnitude with respect to all baselines, all the while delivering superior performance.

For preliminary particulars—among them the loss and advantage-calculation procedures governing antecedent algorithms such as GRPO, DAPO, and GSPO—the reader is referred to Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

The clipping of the probability ratio constitutes a widely adopted expedient in reinforcement learning, its purpose being to stabilize policy updates and to forestall the large, destabilizing parameter shifts that would otherwise attend optimization. The mechanism assumes heightened importance whenever importance sampling is deployed in off-policy settings. When importance sampling is reweighted by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$ is indeed recoverable; the variance of such an estimate, however, may be inflated to a considerable degree, as characterized in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

In order to temper this inflation of variance, prior works (e.g., Schulman et al. (2017)) confine the probability ratio within a fixed clipping bound parameterized by the hyperparameter ϵ , as stipulated in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, curtails the model’s exploration of the diversity space, and it does so most acutely in those regions to which the old policy assigns scant probability mass. Under such circumstances the absolute bound governing diminishing probabilities is inherently narrow, so that the capacity for substantive policy updates is correspondingly restricted. Although methods of the DAPO variety endeavor to alleviate this difficulty by way of asymmetric clipping bounds, the fundamental limitation nonetheless endures. GSPO, for its part, discards entire responses whenever the sequence-level importance ratio transgresses the asymmetric bounds—thereby squandering the more informative tokens while retaining a preponderance of high-variance ones, a combination that in turn engenders training instability.

Viewed through the lens of variance–bias control, a more serviceable approach is to impose constraints that are themselves probability-dependent, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3), and safeguarding the non-negativity of the radicand by taking its maximum with zero, we arrive at a closed-form solution for $r(x)$. What emerges is a family of **dynamic-adaptive clipping bounds** that vary in concert with $q(x)$ and with the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping scheme thereby licenses a greater measure of policy exploration precisely within the low-probability regions. Furthermore, drawing inspiration from the dual clipping method of (Ye et al., 2020), we fix the maximum bounds for both positive and negative advantages at 10, a precaution that guards against the overtraining that excessively permissive clipping bounds would otherwise invite. For the detailed derivation and the attendant analysis of these dynamic clipping bounds within the importance-sampling framework, the reader is directed to Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In antecedent works—GRPO, DAPO, and GSPO among them—the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing solely the rewards $R_{j=1,\dots,G}^i$ elicited for the same prompt at the i -th step, in the manner of Equation (11) in Appendix A.3. Such a procedure, however, is liable to several infelicities. First, whenever the stochasticity of response sampling renders all rewards identical at a given step, the advantage collapses to zero, and the prompt is thereby debarred from contributing to parameter updates—this notwithstanding the potentially valuable divergences among the underlying reasoning trajectories. Second, the randomness inherent in high-entropy sampling can beget markedly skewed label counts, which in turn provoke large fluctuations of the standardized advantage values across steps—fluctuations severe enough, on occasion, to reverse their sign and so to destabilize training.

Concurrently, recent work (Yue et al., 2025) intimates that RLVR chiefly refines a policy by modulating the likelihood of correct samples, rather than fundamentally recasting the pre-existing distribution. In light of these considerations, the aggregate reward distribution of the responses to a given prompt may reasonably be regarded as being drawn from one and the same global distribution over the entire course of training—a premise that motivates a *cumulative* standardization of the form given in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics in question are computed over the entirety of the responses generated thus far for the same prompt.

To attenuate the fluctuations that beset both the step-specific standardization ($\hat{A}_{new,j}^i$) and the cumulative standardization ($\hat{A}_{total,j}^i$), we introduce a pair of smoothing functions, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each of which constitutes a weighted average of the two standardization schemes whose relative weights evolve with the step index i , as set out in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

As training advances, $\hat{S}A_{new,j}^i$ progressively concentrates its weight upon the current distribution, thereby evincing a preference for accommodating the most recent policy outputs. Conversely, $\hat{S}A_{total,j}^i$ shifts its weight toward the aggregate cumulative distribution, construing the deviation of the current step as a fine-tuning adjustment situated within the broader distribution. Our final advantage $\hat{A}_j^i$ is, in turn, defined as whichever of the two smoothed advantages possesses the smaller absolute value—a choice calculated to diminish the impact that the respective fluctuations of the cumulative and current-step standardizations would exert upon training stability, as codified in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

Once a prompt has entered into the model optimization, the responses generated at every subsequent step continue to partake of that optimization, irrespective of whether the current advantage happens to be zero. In consequence, even when identical rewards materialize within a step, the corresponding responses still update the model by way of the advantage $\frac{1}{i} \hat{A}_{total,j}^i$, so that useful learning signals are preserved and data efficiency is thereby improved.

3.1 OTM LOSS

Although the **Token-Level Mean loss (TLM)** (the loss Equation (13) employed in DAPO) equilibrates the contributions of all tokens, it does so at the cost of disrupting the relative relationship that ought to obtain among responses of differing length. Consider, by way of illustration, two responses to a common prompt: response A , whose advantage is 1 and whose length is 500, and response B , whose advantage is 0.5 and whose length is 1500. Under TLM the loss for A is weighted by $\frac{500}{500+1500} = 0.25$ and that for B by $\frac{1500}{500+1500} = 0.75$; the total loss contribution of A therefore amounts to $1 \times 0.25 = 0.25$ and that of B to $0.5 \times 0.75 = 0.375$. The upshot is that B comes to exert the greater influence (0.375) notwithstanding its inferior advantage (0.5).

By the same token, when the average is taken across batches, the **Sequence-Level Mean loss (SLM)** (the loss Equation (12) employed in GRPO) enfeebls the magnitude of the advantage—an operation tantamount to scaling it down by a factor of G once the advantages have been computed. When the batch size is large in particular, this has the effect of dismantling the correlational structure that the original standardized advantages had established. We contend, on the contrary, that standardized advantages already encode relative relationships across responses, and that averaging over all tokens within a single response by $|o_i|$ not only conserves that original relationship but additionally apportions importance equitably among the constituent tokens of the current response. Accordingly, we average only over the tokens internal to a single response, eschewing any averaging across the responses of a shared batch—a formulation we designate the **Only Token Mean loss (OTM)**, given in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As is the case in DAPO, the DCPO loss is left unconstrained by the KL divergence. For the lower and upper clipping bounds alike, we adopt the dynamic clipping bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets Our experiments were carried out over four distinct models—Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b)—and drew upon a merged training corpus of approximately 25k mathematical problems, itself constituted by uniting the DAPO-Math-17K corpus (Yu et al., 2025) with levels 3–5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). Every problem was rendered through the Qwen-Math template (see Section A.8), which we applied uniformly to both training and evaluation.

Evaluation We appraised our models against four widely used mathematical-reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). In the case of **MATH500**, we confine our reporting to Avg@1, owing to its comparatively large size (500 problems). For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) datasets, we report both Avg@1 and Avg@32, the better to secure a robust and comprehensive evaluation.

- **Avg@1**: this metric denotes the standard accuracy realized under greedy decoding, and thus gauges the performance of the model’s single best prediction.
- **Avg@32**: this metric computes the mean accuracy over 32 responses sampled per problem at a temperature of 1.0 and a top-p of 1.0, thereby affording insight into the robustness and stability of the trained policy distribution.

Baseline Settings We adopted GRPO and DAPO as baselines and, for Qwen2.5-Math-7B and Qwen2.5-14B, augmented this set with GSPO as a further point of comparison. The remaining settings—among them the several clipping thresholds ϵ_* , the codebase, and the other common training hyperparameters—are documented in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 consolidates the performance of a range of preference-optimization methods—GRPO, DAPO, GSPO, and our proposed DCPO—across the four mathematical-reasoning benchmarks MATH500, AMC23, AIME24, and AIME25. By reporting both the greedy-decoding results (Avg@1) and the sampling-based results (Avg@32), we furnish a comprehensive appraisal of each method’s robustness and generalization. At every model scale, DCPO consistently manifests performance that is either superior to, or on a par with, that of GRPO, DAPO, and—where available—GSPO, a pattern that attests to its efficacy as a general-purpose preference-optimization technique within the reinforcement learning of LLMs.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO attains the highest average scores (47.2/32.8), thereby outpacing GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO yields (37.6/22.7), exceeding GRPO (+1.3/+1.7) while remaining competitive with DAPO—registering a notable gain in Avg@1 (+1.0) and conceding only a marginal shortfall in Avg@32 (-0.4). For Qwen2.5-Math-7B, DCPO attains (57.1/45.2), surpassing GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9) alike. On the largest model under test, Qwen2.5-14B, DCPO delivers a compelling result of (53.2/39.0), affording substantial improvements over GRPO (+6.5/+7.7) while outperforming DAPO by (+1.3/+0.1) and GSPO by (+4.2/+5.5) with respect to Avg@1 and Avg@32. Collectively, these results throw into relief the robustness of DCPO’s improvements across disparate model capacities, together with its superior alignment to the reward signal under both the greedy and the stochastic decoding paradigms.

Robustness on AIME under Sampling (Avg@32) A particularly noteworthy trend surfaces on the AIME24 and AIME25 datasets—high-difficulty competition-mathematics benchmarks expressly devised to stress-test a model’s reasoning faculties. Under Avg@32, which interrogates the model’s capacity to produce correct responses amid high-entropy sampling, DCPO evinces conspicuously heightened robustness at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B attains 38.8, outstripping GRPO (32.1), DAPO (31.6), and GSPO (34.9), whereas on AIME25-Avg@32, DCPO-14B reaches 19.0, surpassing GRPO (10.5), DAPO (15.3), and GSPO (9.9). Such gains intimate that DCPO not merely elevates average-case performance but additionally enriches the diversity

Table 1: Performance across benchmarks. Avg@1 corresponds to greedy decoding, whereas Avg@32 corresponds to 32-fold sampling. Boldface designates the best value attained, and Δ quantifies the improvement realized over the respective baselines.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

and reliability of its correct generations—a consideration of especial import under sampling-based decoding, where beam search or temperature-driven variability holds sway. That the method contrives to optimize reward fidelity and exploration simultaneously under high-entropy decoding underscores its practical worth in real-world settings, wherein a plurality of plausible outputs is habitually sampled.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio** (TCR) as the proportion of tokens that, by virtue of clipping, are excluded from the policy updates effected during back-propagation. As Figure 1 makes plain, the behavior of the TCR diverges markedly from one reinforcement learning method to the next.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N the number of microbatches comprising a single training step.

Under GRPO training, the TCR traces trajectories that diverge according to model size. For the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) the TCR mounts with successive training

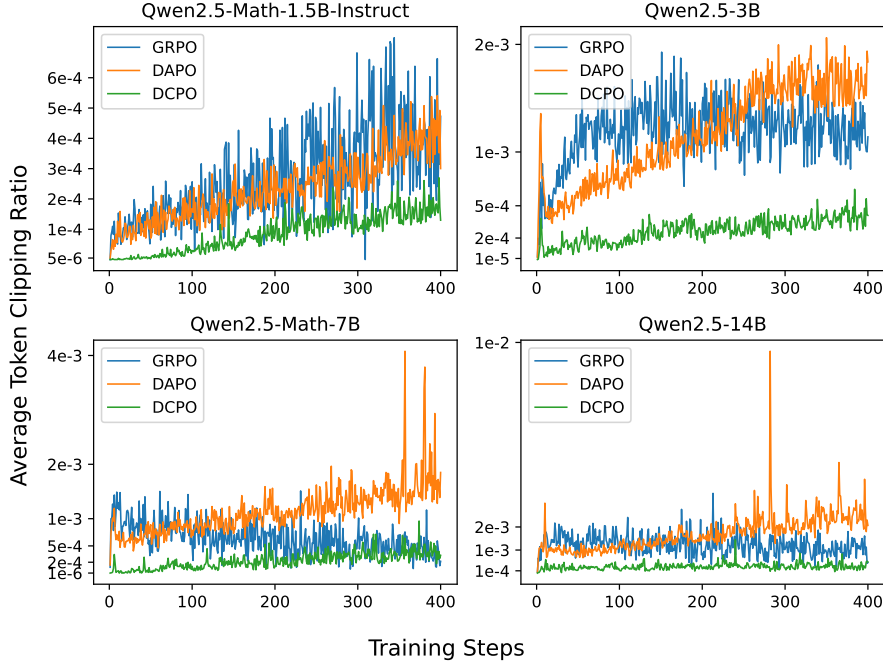


Figure 1: The TCR as it varies across the several models and methods.

steps, whereas for the larger models (Qwen2.5-Math-7B and Qwen2.5-14B) it gradually subsides. DAPO, by contrast, exhibits a uniformly ascending TCR trajectory across all models—an indication that, as DAPO training unfolds, an ever-growing contingent of tokens is clipped, so that a heightened proportion of policy updates comes to rest upon partial or truncated responses. Both GRPO and DAPO, moreover, betray pronounced fluctuations and sporadic anomalous spikes in the TCR , disturbances liable to inject instability into the optimization process and, potentially, to precipitate entropy collapse. In parallel, the average TCR of GSPO exceeds 11% on Qwen2.5-Math-7B and surpasses 15% on Qwen2.5-14B—values far in excess of the average TCR exhibited by token-level clipping methods, and ones that squander a greater quantity of informative tokens.

DCPO, by contrast, sustains a remarkably stable and efficient training regime. Irrespective of model size or training stage, its TCR remains comparatively constant and an order of magnitude below that of GRPO and DAPO. This lower and steadier TCR signifies that DCPO discards fewer tokens, thereby permitting a larger share of the complete response sequences to inform the policy updates and liberating additional latitude for the model to explore the diversity of rare tokens. The finding foregrounds DCPO’s superior sample efficiency relative to the baselines, achieved even as it holds the scope for outlier variation firmly in check.

Taking Qwen2.5-Math-7B as our illustrative case, we observe that the token distribution rapidly concentrates upon high-confidence tokens: after roughly 60 steps, some 95% of generated tokens carry a probability ($q(x) > 0.9$); this fraction ascends to approximately 97% after roughly 100 steps and continues to climb through the later stages of training. As Figure 4d illustrates, once the old probability satisfies ($q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$), the new probability $p(x)$ participates in the model updates within the interval $[\frac{1}{1+\epsilon}, 1]$ and escapes discarding under any clipping method whatsoever. It follows that the overwhelming majority of tokens reside in a region that is never clipped—a circumstance that accounts for the low overall TCR witnessed in practice. Meanwhile, Figure 1 reveals that the TCR swells during the later phases of most training runs, signalling that a progressively larger proportion of low-probability tokens may be discarded and, in consequence, that the model’s capacity to explore the diversity of rare tokens is curtailed.

Prior work (Wang et al., 2025) has demonstrated that **high-entropy tokens**—which are, precisely, the low-probability ones—constitute the principal engines behind the emergence of the model’s reasoning capabilities. Because entropy and probability stand in inverse relation, the rarer a token, the

greater the informational content it conveys. Furnishing a more capacious admissible update range for such low-probability tokens therefore enlarges the model’s effective exploration space and, in turn, facilitates the acquisition of diverse reasoning capabilities.

These empirical and theoretical findings jointly vindicate our proposed **dynamic-adaptive clipping** scheme, which relaxes clipping for tokens satisfying $q(x) < \frac{1}{1+\epsilon}$ while conserving the safe interval reserved for high-confidence tokens. By adaptively dilating the update interval for the high-entropy tail of the distribution, the method restores the model’s ability to learn from rare and informative tokens and, as a corollary, enhances its reasoning ability.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

Throughout RLVR training, a policy update is contingent upon the presence of a nonzero gradient for the response in question. We quantify the efficiency of this process by means of a metric we term the **Response Utilization Ratio** (*RUR*)—namely, the percentage of generated responses that, possessing a nonzero advantage, actually partake of the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 makes it apparent that the overall average *RUR* for responses generated under GRPO training is low, its mean amounting to a mere 44.6% across the four models. The implication is that upwards of half of the generated responses are discarded—a fact that throws into relief the pronounced inefficiency of response utilization during the exploration of diversity. Our proposed DCPO method, by contrast, attains an average *RUR* of approximately 70% once the first epoch has elapsed. This constitutes a substantial improvement, corresponding to an absolute increase of 28% and a relative gain of 64% over that of GRPO.

Table 2: The average *RUR* computed over the course of 400 training steps.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 demonstrates that the *RUR* under GRPO training deteriorates substantially over time, and that it does so irrespective of a model’s size or type. Strikingly, the *RUR* plunges from above 90% to below 50%; for Qwen2.5-Math-7B, to take one instance, it sinks as low as 30% after approximately 200 training steps. This finding suggests that GRPO’s reliance upon current-step advantage standardization begets a substantial fraction of zero gradients, and hence that the model’s exploration of diversity grows conspicuously inefficient in the later stages of training. Our proposed DCPO method, which instead deploys a smooth advantage-standardization scheme, sustains a consistently high *RUR*: once the first epoch concludes, the *RUR* swiftly stabilizes in the vicinity of 70% and continues to improve as training proceeds.

We observed the *RUR* to exceed 90% during the first several steps yet, from the second epoch through to the end of training, to hover around 70% without ever attaining 100%. In what follows we anatomize the reasons behind these phenomena and, in so doing, expose the latent value of the SAS mechanism.

First, the initial *RUR* in excess of 90% is attributable in large measure to the outputs of the early-stage model, which fail to observe the requisite formatting and thereby incur negative rewards that artificially inflate the *RUR*. This indicates that models of various kinds are able to assimilate fixed answer patterns with dispatch, so that subsequent training ought to redirect its emphasis toward the exploration of the model’s reasoning capabilities. Second, our analysis discloses two principal factors conducive to a persistently zero advantage—and hence to the *RUR* stabilizing around 70% rather than reaching 100%. **(i)** Nearly half of the mathematical problems are sufficiently elementary that the model solves them correctly without fail, and their exclusion from policy updates is warranted precisely because the model’s core competencies stand in no need of further refinement in these cases. **(ii)** The remaining instances correspond to responses that are consistently erroneous, a condition that may arise from either of two sources. On the one hand, certain problems prove too formidable

for the current model to solve, even when furnished with a correct ground-truth label. On the other hand, the DAPO-Math-17K dataset harbors a number of items whose labels have been converted into mislabeled integers, thereby introducing spurious “incorrect” labels. In such cases, the discarding of zero-advantage responses operates as a species of filtering, mitigating the hazard of updating the model on the basis of noisy or mislabeled data.

Unlike GRPO and DCPO, DAPO expels from the generated data those responses that share identical labels, a practice that renders any direct comparison of *RUR* problematic. We therefore gauged DAPO’s training efficiency by contrasting the number of generated responses and the training time it requires to complete the same number of update steps. To offset the low sampling efficiency occasioned by GRPO’s meagre *RUR*, we fixed the generation batch size at between one and four times the training batch size and adjusted it dynamically over the course of DAPO training. The experiments reveal that, across the four models, DAPO demands between three and five times as many generated responses as DCPO to accomplish an equivalent number of parameter updates—at least doubling the training GPU-hours while yielding no superior performance. We observed, furthermore, that DAPO’s data-utilization efficiency deteriorates as model size grows: on Qwen2.5-14B, for example, the proportion of responses that DAPO actually employs fell below 30% after roughly 300 steps. This attests that DAPO’s dynamic-sampling regime suffers from low data utilization and training efficiency, and that it forfeits opportunities to explore the model’s diversity.

In sum, DCPO represents a signal advance over both GRPO and DAPO: by substantially enhancing response utilization and training efficiency—harnessing the sampled data in full so as to endow the model with more expansive room for the exploration of diversity—it delivers commensurate improvements in performance.

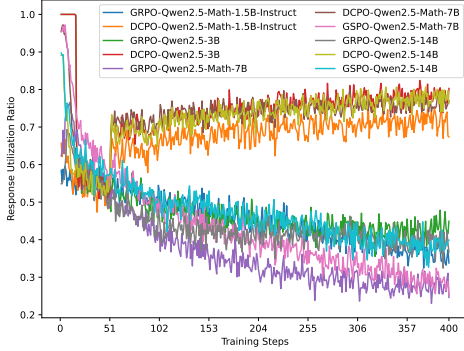


Figure 2: The evolution of the *RUR* over the course of training.

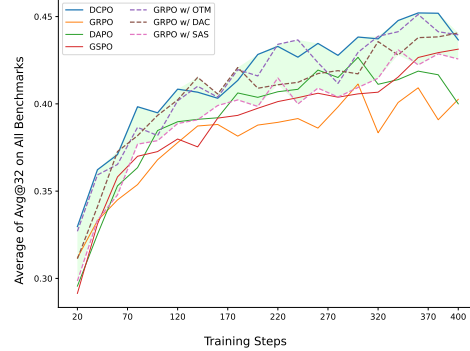


Figure 3: An ablation conducted with the average Avg@32 metric on Qwen2.5-Math-7B.

5.4 ABLATION STUDIES

To assess the contribution rendered by each constituent of DCPO, we conducted an ablation study on Qwen2.5-Math-7B, taking **Avg@32** as the evaluation metric—a choice that foregrounds the robustness and stability of the learned policy distribution. In the interest of fairness, each experiment perturbs a single component of the baseline GRPO framework while holding all remaining settings identical and excising the KL-divergence term, so as to bring the configuration into alignment with DAPO, GSPO, and the full DCPO. The learning curves spanning steps 20 through 400 are displayed in Figure 3.

The learning curves associated with the three individual components reveal that the model steadily refines its inferential ability, coming to surpass GRPO and to match—or indeed exceed—the performance of DAPO and GSPO. Each modification thus confers a distinct benefit, and their conjunction yields a substantial aggregate gain over the baselines, thereby evidencing the efficacy of the DCPO pipeline relative to those baselines in reinforcement learning tasks. In the later stages of training the baselines plateau and cease to improve, whereas DCPO continues to trend upward, both at the level of the individual component and in the aggregate. The ablation studies corroborate that every component contributes favorably to training efficiency, and that their synergistic integration issues in cumulative gains and enhanced stability. The detailed performance is set out below.

(i) **Only-Token-Mean loss (OTM)**: when GRPO aggregates the loss within a single response rather than across the entire batch, the resulting model attains a performance that eclipses the baselines, which are beset by lower data utilization and erratic performance. (ii) **Smoothed Advantage Standardization (SAS)**: this component supplants the conventional stepwise advantage with the cumulative smoothed advantage, appreciably improving upon GRPO and approaching the performance of DAPO. The gain demonstrates that aggregating reward statistics across training steps stabilizes optimization and elevates final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: substituting a probabilistically adaptive clipping boundary for GRPO’s fixed one yields marked improvements. DAC in isolation surpasses GRPO, DAPO, and GSPO, and proves more stable than the employment of OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: conjoining these three mechanisms secures the finest and most stable overall results, decisively outperforming GRPO, DAPO, and GSPO. The synergy obtaining among OTM, SAS, and DAC demonstrates that each design choice contributes uniquely to data-efficient and stable reinforcement learning, thereby improving both the performance and the robustness of large language models.

Taken as a whole, the ablation study confirms that each component of DCPO contributes favorably to overall performance and that their combination gives rise to substantial cumulative gains. The results thus validate the efficacy of the proposed mechanisms in advancing data efficiency and stability within the reinforcement learning of LLMs.

6 CONCLUSION

In the present work we have sought to remedy two cardinal limitations of extant policy-optimization methods: the restricted token-level exploration that fixed clipping imposes within low-probability regions, and the markedly diminished sample efficiency that follows from standardizing rewards at the current step alone. To this end, we introduced **Dynamic Clipping Policy Optimization (DCPO)**, a novel reinforcement learning pipeline that augments the reasoning capabilities of large language models by way of two central innovations: (i) a **dynamically adaptive token-level clipping** mechanism that recalibrates its bounds in accordance with the old-policy probabilities, thereby enabling more efficient exploration of rare, low-probability tokens; and (ii) a **Cumulative-Smoothed Advantage Standardization** that aggregates the cumulative reward distribution across optimization steps, thereby tempering the reward randomness engendered by high-entropy sampling. Extensive experiments spanning four mathematical-reasoning benchmarks and four models attest to DCPO’s efficacy. The DCPO-7B variant, for instance, attained an Avg@32 score of 38.8 on the demanding AIME24 benchmark, thereby outperforming GRPO (32.1) and DAPO (31.6). DCPO further enlarged the response-utilization ratio by 28% relative to GRPO, doubled the training efficiency exhibited by DAPO, and compressed the token clipping ratio by an order of magnitude. These results throw into relief DCPO’s capacity to markedly improve data utilization and the exploration of diversity within RLVR. Future work will pursue the extension of DCPO to further domains, code generation and semantic reasoning foremost among them.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In antecedent works, GRPO and DAPO included, the advantage $\hat{A}_{j,t}^i$ for the token t in the response j is computed by standardizing the reward R_j^i against the mean μ^i and the standard deviation σ^i of the rewards borne by the G responses generated at the i -th step, in the manner set out in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

Whenever the rewards elicited for a given prompt prove identical, the responses whose advantages thereby vanish make no contribution to the model update, and are in effect squandered.

A.3.1 GRPO

Shao et al. (2024) advanced **Group Relative Policy Optimization (GRPO)**, which first samples G responses for each query, assigns rewards R by way of a rule-based reward function, and estimates token-level advantages in the manner of Equation (11). The model parameters are subsequently updated by means of Equation (12), an objective designated the **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (12)$$

The first term averages the token-level advantages over all tokens within a response and thereafter over all G responses within the batch. The $\mathbb{D}_{\text{KL}}$ penalty restrains the policy from straying too far from a reference policy (e.g., a pre-trained model), while the fixed clipping threshold ε forestalls inordinately large updates by delimiting the change permitted in the token probability ratios. This averaging strategy, however widely adopted, inherently dilutes the importance attaching to each individual response.

A.3.2 DAPO

In a kindred vein, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, having observed that, under GRPO, longer sequences exert a disproportionate sway over the gradient update. They standardize the token counts across the entire batch, an operation that gives rise to the **Token Level Mean loss (TLM)** of Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. The procedure discards those prompts whose responses uniformly share an identical reward and regenerates responses so as to preserve the batch size—a strategy that engenders severe data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, contending that token-level clipping is apt to inject high-variance noise into the gradient estimates—noise that tends to accrue over long sequences and to be further magnified by the clipping operation itself. GSPO ameliorates this difficulty by supplanting token-wise clipping with sequence-wise clipping, whereby a more stable and more effective training signal is obtained.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_\theta(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)} \right) \right)$$

GSPO nonetheless discards upward of 10% of the responses that carry non-zero advantages, thereby incurring considerable data waste at the expense of training efficiency. GRPO, for its part, though it discards responses exhibiting sequence-level high variance, still retains a large contingent of responses characterized by token-level high entropy—a retention that occasions training instability.

A.4 REWARD CALCULATION

In order to furnish a supervision signal that is at once verifiable and interpretable, we adopt a rule-based reward model that adjudicates each generated response on the twin criteria of answer correctness and format compliance, as codified in Equation (15). Specifically, a reward of +1 is conferred when both the format and the answer are correct; a reward of 0 when the format is correct but the answer is not; and a reward of −1 when the format is itself incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response group for a given problem at the i -th training step, and the reward is computed on the basis of the response’s format and correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

The clipping of the probability ratio constitutes a crucial technique in reinforcement learning, serving to stabilize policy optimization and to avert the large, destabilizing parameter updates that would otherwise ensue. The mechanism assumes particular significance whenever importance sampling is enlisted to estimate expected values under a new policy from samples that an old policy has generated.

The expected value of a function $f(x)$ under the new probability $p(x)$ admits of rewriting as an expectation under the old probability $q(x)$, mediated by the importance-sampling weight, as exhibited in equation 16:

$$\int f(x)p(x) dx = \int f(x)\frac{p(x)}{q(x)}q(x) dx = \mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance is liable to considerable inflation—a difficulty endemic to importance sampling. The true variance of the function $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x)\frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The disparity between the importance-sampled variance and the true variance lays bare the potential for inflation, as displayed in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x)\frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To attenuate this disparity, a great many prior works—among them PPO Schulman et al. (2017) and various RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025)—impose a fixed symmetric clipping bound upon the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently constrains the policy’s capacity for exploration—a constraint that proves especially troublesome in regions to which the old policy assigns a low probability mass. In such cases the absolute bound governing low probabilities is severely circumscribed, so that the potential for meaningful policy updates is correspondingly restricted. And while methods of the DAPO variety endeavor to redress this by introducing asymmetric clipping bounds, the fundamental limitation of a non-adaptive constraint persists undiminished.

In addition, recent studies such as Yue et al. (2025) intimate that the reasoning paths harnessed by RLVR models reside, for the most part, within the sampling distribution of the base model—a proposition further borne out by the minimal parameter shift discernible during off-policy batch updates, itself reflected in the small fraction of parameters that are clipped. Given this stability, we propose a more serviceable alternative for constraining the probability ratio $r(x)$: a dynamic-adaptive mechanism that incorporates the probability directly into the restriction, as in Equation (21), thereby supplanting the conventional fixed-clipping approach of Equation (20).

$$|(\frac{p(x)}{q(x)} - 1)p(x)| \leq \epsilon \quad (21)$$

Expanding the substitution $p(x) = r(x)q(x)$ within the inequality of the foregoing equation yields the expression given in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to remain mathematically valid and meaningful, we must abide by the fundamental constraints governing probability distributions and their ratios. Specifically, as Equation (23) indicates, the probabilities $p(x)$ and $q(x)$ must be non-negative and no greater than 1; it follows in turn that the probability ratio $r(x)$ must likewise be non-negative, as must the clipping hyperparameter ϵ .

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To guarantee the validity and the practical applicability of the clipping bound derived from Equation (22), we stipulate the necessary conditions upon its bounds as delineated in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Proceeding from Equation (22) to Equation (24), we find that the probability ratio $r(x)$ ought to satisfy the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we consider the overlapping region centered upon the ratio value of 1, so as at once to minimize the discrepancy of the bounds between the two conditions and to preserve consistency with the conventional fixed bound of GRPO; on this basis we define the following conditions to ensure the validity of the clipping bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, and in order to guarantee that the radicand remains no less than zero, we adopt the expedient of comparing with 0 and retaining the maximum, in the manner of the approximation $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. We thereby arrive at the following conditions, which constitute the **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower clipping bounds appearing in Inequality 27, we assign distinct upper and lower clipping-threshold hyperparameters, ϵ_{high} and ϵ_{low} , in order to align the clipping bound with that of GRPO at specific points.

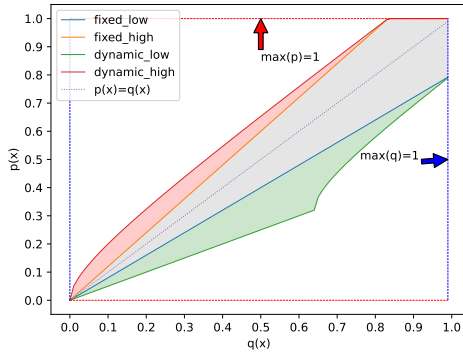
A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

In order to ensure that our dynamic-adaptive clipping bounds coincide with the fixed symmetric clipping bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as depicted in Figure 4d, we derive the specific values of ϵ_{low} and ϵ_{high} by solving the equations set out in Equation (28).

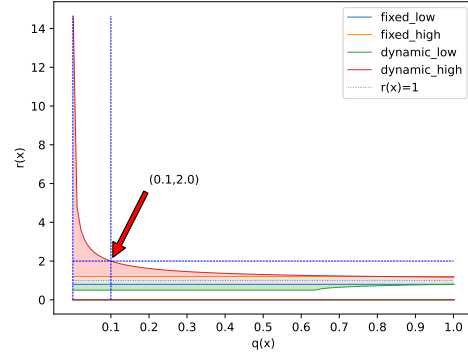
$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2}\sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

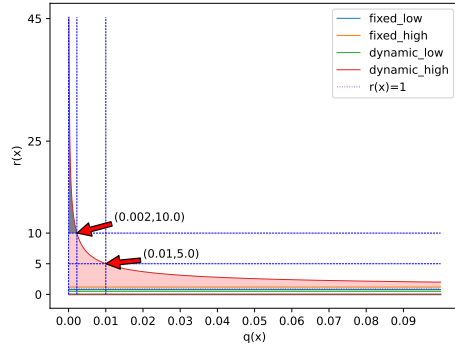
In prevailing practice (e.g., GRPO) the fixed clipping threshold ϵ is customarily set to 0.2, whence our clipping hyperparameters emerge as approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. The inequality $\epsilon_{low} \neq \epsilon_{high}$ here serves a purpose altogether distinct from that of the asymmetric fixed-clipping setting employed in DAPO.



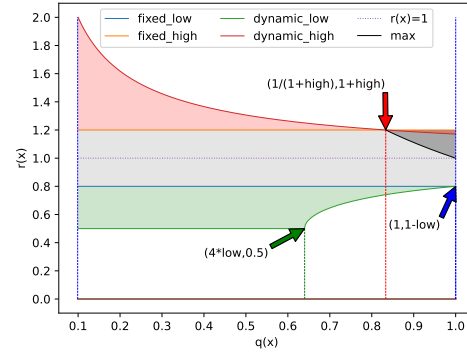
(a) The regions of $p(x)$ that escape clipping.



(b) $r(x)$ bounds ($q(x) \in [0, 1]$).



(c) $r(x)$ bounds ($q(x) \in [0, 0.1]$).



(d) $r(x)$ bounds ($q(x) \in [0.1, 1]$).

Figure 4: A comparison of clipping bounds. The lines trace the bounds for fixed clipping ($\epsilon = 0.2$) as against those for dynamic-adaptive clipping ($\epsilon_{low} = 0.16$, $\epsilon_{high} = 0.2$).

Dynamic versus fixed clipping: Figure 4 illustrates the comportment of the bound under the fixed clipping scheme adopted by GRPO ($[1 - \epsilon, 1 + \epsilon]$) and under our probability-aware adaptive clipping. For a token with an old probability $q(x)$, the fixed bound furnishes a constant relative clipping width of 2ϵ ; in consequence, as $q(x)$ diminishes, the absolute exploration space it permits contracts linearly (Figure 4a). The relative width of the dynamic adaptation, by contrast, is inversely proportional to $q(x)$, so that the exploration space is held approximately constant throughout the probability range.

Effect of decreasing $q(x)$: When $q(x)$ grows small, the admissible interval $[1 - \varepsilon, 1 + \varepsilon]$ of the fixed clipping scheme, being independent of $q(x)$, affords the model no enhanced capacity to explore low-probability tokens; the dynamic-adaptive bound, on the other hand, expands rapidly (Figure 4b), thereby conferring a larger relative space upon rare tokens—precisely the behavior that efficient RL requires.

Behavior for extremely low probabilities: Figure 4c depicts the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the upper bound on $r(x)$ under dynamic-adaptive clipping increases rapidly, in inverse proportion to the square root, whereas the corresponding value under fixed clipping remains constant. Drawing inspiration from the dual-clipping strategy of (Ye et al., 2020), we enforce a hard ceiling $r_{\max} = 10$ so as to preclude the attendant instabilities (gradient vanishing, explosion, and the like). The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, throughout which the upper limit of adaptive clipping is pinned to $r_{\max}$.

Upper and lower bounds: As regards the lower bound, when $q(x) \leq 4\varepsilon$ the dynamic-adaptive clipping yields $dynamic_{low} = 0.5$, whereas fixed clipping retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$; once $q(x)$ ascends beyond this threshold, the dynamic-adaptive lower bound contracts smoothly toward the fixed value. In like fashion, the dynamic-adaptive upper bound exceeds the fixed bound whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$ and coalesces with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \varepsilon_{low}), dynamic_{high}(q(x), \varepsilon_{high})]$ thus effects a probability-dependent widening of the admissible region for rare tokens while conserving the tightness of the fixed interval for common ones (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) comprises both a positive and a negative advantage term. Under the minimum-loss operation, the ratio $r(x)$ associated with a positive advantage is clipped to the interval $[0, dynamic_{high}(q(x), \varepsilon_{high})]$, whereas that associated with a negative advantage occupies the interval $[dynamic_{low}(q(x), \varepsilon_{low}), \infty)$. Because an excessively large $r(x)$ is apt to destabilize training, we cap the ratio for both positive and negative advantages at $r_{\max} = 10$ —a precaution inspired by Ye et al. (2020) that ensures the adaptive limits do not give rise to inordinately high importance weights.

In short, whether the advantage be positive or negative, dynamic-adaptive clipping—unlike its fixed counterpart—endows the model with greater room to explore diversity whenever rare tokens possess a low probability.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration, at which prompts are enlisted to generate responses and to update the model parameters. We define the following statistics in the service of reward normalization.

For the mean of the response rewards, we specify the corresponding group of functions as in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

For the variance of the response rewards, we specify the corresponding group of functions as in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^{i-2} + (i-1) \sigma_{old}^{i-2} + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To curtail resource consumption, we compute the values of σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29) by way of equivalent expressions in μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i , rather than by direct computation over the original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

Throughout the experiments reported in this paper we employ the following template, itself derived from the Qwen-Math template (Yang et al., 2024b).

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We fixed a symmetric clipping threshold of $\epsilon = 0.2$ for GRPO; for DAPO we adopted lower/upper clipping thresholds of $(0.2, 0.28)$, supplemented by a 512-token soft-punish cache and a maximum probability ratio of 10 for the negative advantage; for GSPO we set lower/upper clipping thresholds of $(3e - 4, 4e - 4)$; and for DCPO we employed low/high clipping thresholds of $(0.16, 0.2)$, thereby preserving the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ as GRPO (Figure 4d), and we capped the maximum probability ratio at 10 for both positive and negative advantages—a provision that departs from DAPO.

Training proceeded for 400 steps with a batch size of 512 for response generation and a mini-batch size of 32 for parameter updates. At a temperature of 1 and a top.p of 1 we generated $G=16$ responses per problem, while the maximum prompt length was set to 1,024 and the maximum response length to 3,072—the latter far below the 20k maximum employed in the original DAPO.

Our training codebase was adapted from Verl (Sheng et al., 2024) and run on $32 \times H20$ GPUs. Save where noted otherwise, the hyperparameters were matched to those of DAPO. For the loss computation we retained the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation of GRPO or DAPO, in preference to the approximate weighted calculation used in Verl, so as to guarantee compatibility across differing microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 portrays the model’s performance in terms of the average Avg@1 metric—a quantity obtained by averaging the greedy-decoding (Avg@1) results across the MATH500, AMC23, AIME24, and AIME25 benchmarks, and thus representative of the model’s best performance. As the

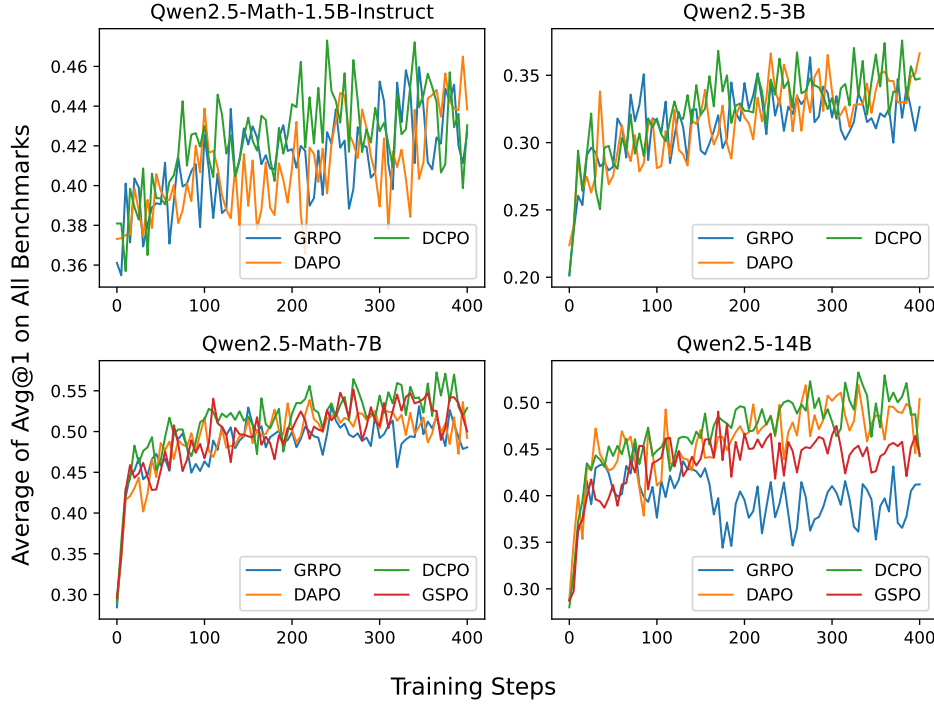


Figure 5: Avg@1 performance across the several benchmarks.

figure attests, DCPO consistently outstripped GRPO and DAPO across all four models, and likewise outpaced GSPO on the two models against which GSPO was evaluated. Its overall performance, moreover, proved more stable than that of the baselines, being untroubled by the pronounced performance drop witnessed under GRPO and the pronounced fluctuations witnessed under DAPO. This suggests that DCPO is able to elicit stronger reasoning capabilities during training, thereby endowing the model with greater latitude to explore still better performance.

Avg@32 Trend Figure 6 juxtaposes the Avg@32 performance of DCPO with that of GRPO, DAPO, and—where available—GSPO across differing model sizes on the AMC23, AIME24, and AIME25 benchmarks, and in so doing renders DCPO’s exceptional robustness and stability unmistakable. GRPO, by contrast, evinces marked fluctuation and occasional outright collapse: under GRPO, the Qwen2.5-14B model even suffers a pronounced performance drop after approximately 170 training steps. Across every evaluated model (from 1.5B to 14B), the DCPO performance trajectory traces a consistently upward course accompanied by minimal fluctuation; even in the earliest stages of training DCPO sustains the swiftest growth rate while preserving its upward trend. Furthermore, in comparison with DAPO, DCPO attains superior performance even as it economizes half the training time and resources. These experimental results together demonstrate the efficacy and robustness of dynamic-adaptive clipping and smooth advantage standardization in enhancing exploration of diversity at both the response level and the label level.

Entropy Trend Figure 7 reveals that GRPO undergoes a precipitous entropy decay in the wake of an initial warm-up phase and thereupon settles rapidly into a flat, low-entropy regime. This entropy collapse betokens a swift forfeiture of policy randomness, which in curtailing the diversity of the sampled responses thereby circumscribes any subsequent policy optimization.

The behavior of DAPO and DCPO, by contrast, is contingent upon the model. In the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO all trace similar declining trajectories; for the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), however, the trends diverge appreciably. DAPO sustains the highest entropy value, exhibiting a distinct upward trend attended by large fluctuations. On Qwen2.5-Math-7B, the entropy of GSPO resembles that of DCPO, and their average-metric performance is likewise comparable; on Qwen2.5-14B, however, GSPO’s

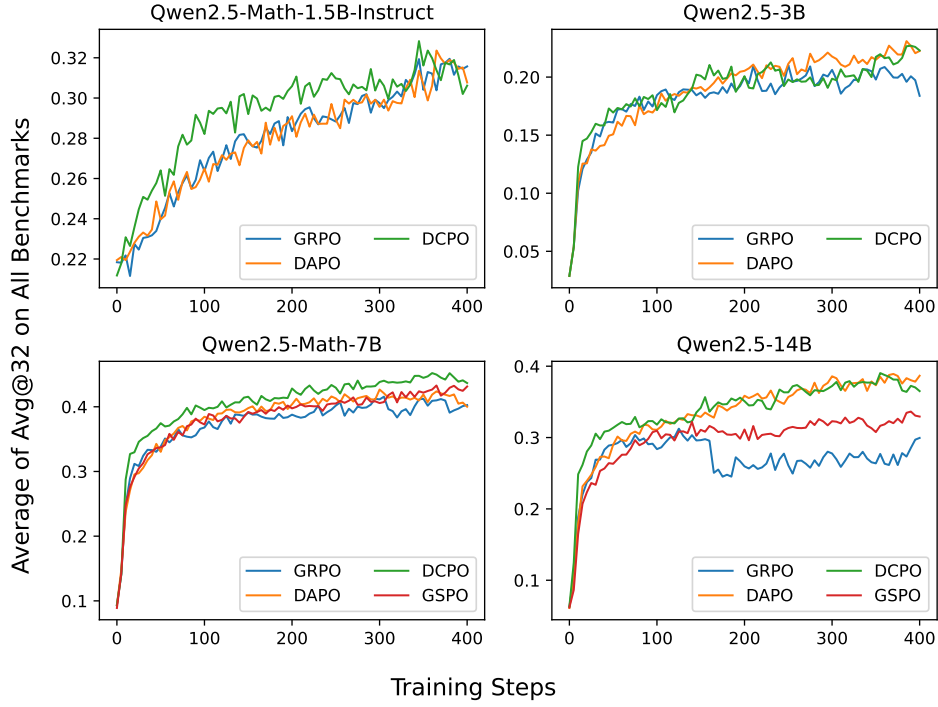


Figure 6: Avg@32 performance across the several benchmarks.

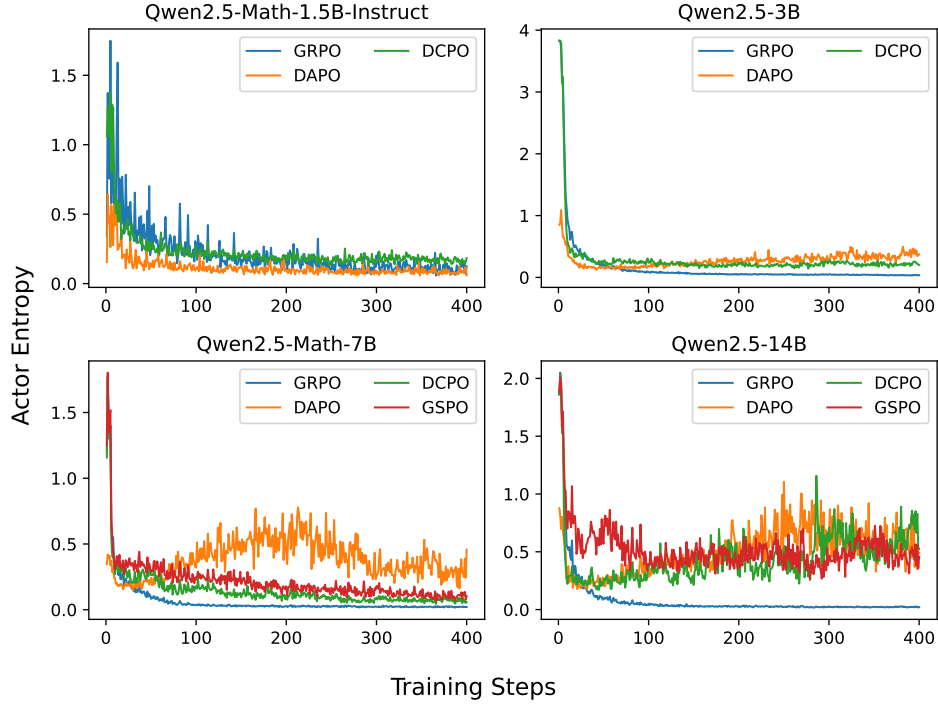


Figure 7: The entropy of the models over the course of training.

performance describes a fluctuating downward trend that culminates in poor results. DCPO, in contradistinction, attains a moderate entropy level: its curve is markedly smoother and remains confined to a bounded band interposed between the low-entropy region of GRPO and the high-variance region of DAPO.

These observations intimate a U-shaped relationship between entropy and training performance. Entropy that is excessively low deprives the learner of exploratory signal and thereby precipitates premature convergence, whereas entropy that is excessively high or unstable compromises training stability. DCPO holds entropy within a moderate and well-balanced range, at once fostering convergence and preserving sufficient exploration—a benefit that grows the more pronounced as model capacity increases. It is therefore plausible that DCPO’s entropy-regulation mechanism underlies its superior inferential performance and enables more robust policy learning across models of varying scale.